Summit B Nayak

Passport: AG236107 | **Work permit:** Indian | **Date of birth:** 15/07/2002 | **Place of birth:** Bhadrak, India | **Nationality:** Indian | **Gender:** Male | **Phone number:** (+91) 9556548618 (Mobile) | **Email address:** summit.b.nayak@gmail.com | **Address:** NISER, Mahanadi Hostel, 752050, Jatni, India (Work) | **Address:** Orada, Tihid, Kolha, 756130, Bhadrak, India (Home)

ABOUT ME

Computational astrophysicist specializing in planetary atmospheres and fully differentiable Bayesian retrieval pipelines. Holding an Integrated M.Sc. in Physics with a minor in Computer Science from NISER, I bridge the gap between numerical radiative transfer modeling and hyperspectral observational data. My core expertise lies in architecting multi-stream forward models and deploying Hamiltonian Monte Carlo (NUTS) sampling via JAX/NumPyro on HPC clusters to resolve severe degeneracies in deep-atmosphere spectra. Currently focused on transitioning these high-resolution frameworks to characterize terrestrial and giant exoplanets using next-generation JWST and VLTI data.

WORK EXPERIENCE

PROJECT STUDENT – NISER : SEPS - DR. GUNESHWAR THANGJAM – 01/05/2025 – 10/08/2025 – BHUBANESWAR, INDIA

- Developed fully differentiable radiative transfer forward models and Bayesian retrieval pipelines (MCMC/HMC) using JAX/NumPyro.
- Optimized highly parallelized (64-core) HPC architectures to resolve severe degeneracies in hyperspectral data.
- Extended legacy Solar System atmospheric pipelines for scalability to exoplanetary datasets (e.g., JWST/VLTI applications).

MASTERS THESIS – NISER : SPS AND SEPS – 01/07/2024 – 22/05/2025 – BHUBANESWAR, INDIA

- Pipeline for extracting surface emission by RTM for venus using VIRTIS-M-IR data
- Fortran, ENVI IDL , Python coding languages
- Radiative Transfer Models
- Atmospheric composition and scattering along with cloud models
- Including ML and AI for future Venus missions

POSTER PRESENTER – IIT ROORKIE : S2 STEP 2025 – 04/02/2025 – 08/02/2025 – ROORKIE, INDIA

- presented my work on surface compositional pipeline for venus from VIRTIS VEx
- attended lectures on planetary sciences' ongoing research by honored national and international speakers
- discussed on future improvement and directions

SEMESTER PRESENTATION – NISER : SCOS : DR SUBHANKAR MISHRA : ADVANCE MACHINE LEARNING – 01/01/2025 – 12/05/2025 – BHUBANESWAR, INDIA

- Modified LSH Algorithm (Fly Olfactory) for Similarity Search
- Coding in Python
- Presentation on same

SEMESTER PRESENTATION – NISER : SCOS : DR SUBHANKAR MISHRA - MACHINE LEARNING – 01/07/2023 – 14/12/2023 – BHUBANESWAR, INDIA

- Lipid Quantification in stained cell images following ImageJ
- Python pipeline for before and after lipid content comparison
- Class Presentation on the same

INTERNSHIP – NISER : SPS - DR. PRASANJIT SAMAL - COMPUTATIONAL DFT – 01/05/2023 – 24/07/2023 – BHUBANESWAR, INDIA

- Density Functional Theory and Computation
- Designing DFT systems with used defined variables
- Coding and simulating in python

INTERNSHIP – IIT MANDI : DR PRASANTH P JOSE - CLASSICAL SYSTEM COMPUTATION – 01/05/2022 – 16/07/2022 – MANDI, INDIA

- simulating solutions to 3D classical problems
- C++ coding and data analysis
- simulation of input based classical environments

SCIENCE TEAM MEMBER – L.I.F.E. (LARGE INTERFEROMETER FOR EXOPLANETS) – 01/08/2026 – Current – REMOTE, INDIA

- Contributing to exoplanet atmospheric characterization models under Dr. Lee Grenfell.

EDUCATION AND TRAINING

01/07/2020 – 01/07/2025 Bhubaneswar, India
MASTERS IN PHYSICS Homi Bhabha National Institute - NISER

Website https://www.niser.ac.in | **Level in EQF** EQF level 7

01/07/2020 – 01/07/2025 Bhubaneswar, India
MINOR IN COMPUTER SCIENCES HBNI - NISER

Website https://www.niser.ac.in | **Level in EQF** EQF level 6

01/04/2018 – 01/04/2020 Berehampur, India
HIGHER SECONDARY EDUCATION IN SCIENCE Khallikote University

Website https://kuu.ac.in | **Level in EQF** EQF level 5

01/08/2026 – 15/08/2026 Bangalore , India
AI/ML FOR GEODATA ANALYTICS IIRS - ISRO

Website https://www.isro.gov.in/IIRS.html | **Field of study** Physics

01/05/2026 – 29/05/2026 Paris, France
ASTROBIOLOGY INTRODUCTORY COURSE RED Astrobiology School

Website https://red-astrobiology.fr/ | **Field of study** Natural sciences, mathematics and statistics

LANGUAGE SKILLS

Mother tongue(s): **HINDI** | **ODIA**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C1	C1	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Parallel computing in HPC environment | Git/Github, Docker, Gitlab | programming (C, C++, Fortran, python, IDL, Matlab,) | Programming: Fortran, C++, Python, bash | LaTeX (very good) | Kali Linux | Data analysis with Python (Numpy, Matplotlib, Pandas, etc.) | MATLAB | data processing and analysis software: Origin (advanced) | C++ | Course on Image classification using Machine Learning and Artificial Intelligence | Python (computer programming) | Microsoft Office | Google Services (Google Maps Google Photo Google Drive) | Social Media

CONFERENCES AND SEMINARS

19/11/2025 – 22/11/2025 Bangalore

ISRO Venus Orbiter Mission (VOM) Conference

Attended technical sessions on pre-flight payload calibration and mission planning.

05/05/2025 – 20/05/2025 Roorkee

IIT Roorkee S2 STEP

Attended Earth and Planetary sciences advancement in Indian research. Presented Venus VIRTIS surface compositional pipeline.

01/07/2020 – 01/07/2025 Bhubaneswar

DISHA Scholarship, Department of Atomic Energy, Govt. of India

Awarded for fundamental scientific research during Integrated M.Sc.